# Lab 02: Software Engineering I

- Software Processes in Action
- Instagram and Spotify
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# 1. Software Process & Activities

- A process is how a team plans, builds, tests, releases, and maintains software.
- Specification: decide what it should do.
- Design/implementation: plan and code it.
- Validation: test it. Evolution: improve it after release.

# 2. Process Models

- Waterfall: planned stages in order.
- Incremental: deliver features in smaller releases.
- Integration/configuration: reuse and connect existing components.
- Plan-driven: emphasizes plans and documentation. Agile: short cycles and feedback.

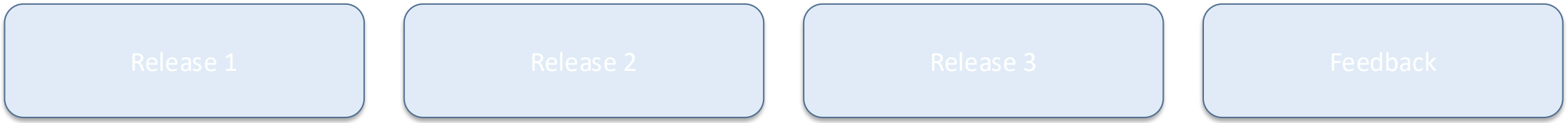
# 3. Waterfall Diagram + Pros/Cons

- Requirements → Design → Implementation → Testing → Deployment → Maintenance
- Pros: clear stages, documentation, easier planning when requirements are stable.
- Cons: late changes can cause expensive rework and delays.



# 4. Incremental Diagram

- Release 1: core features → Release 2: added features → Release 3: improvements
- Feedback helps decide what to build next.
- Smaller releases can make changes easier to handle.



# 5. Integration, Prototypes & Change

- Reuse libraries, frameworks, services, or existing systems.
- Advantage: save time and money. Limitation: components may not fit every need.
- Prototypes test ideas early; change can affect cost, schedule, testing, and documentation.

# 6. Instagram (Meta)

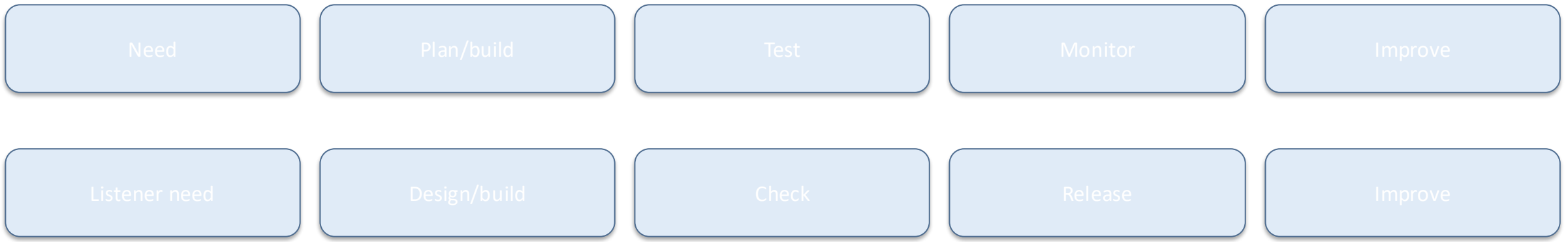
- Instagram supports photo/video sharing, Stories, Reels, messaging, and discovery.
- Meta describes experimentation and automated model launches for Instagram recommendations.
- Interpretation: iterative/experimental practices in these examples; not necessarily every team.

# 7. Spotify

- Spotify streams music and podcasts and supports playlists and discovery.
- Spotify describes cross-functional squads and shared developer tools such as Backstage.
- Interpretation: iterative improvement and reuse in the public examples.

# 8. Application Process Diagrams

- Instagram: need → requirements/design → build model/feature → test/experiment → monitor launch → improve.
- Spotify: listener need → discover/design → build → checks/experiment → release/monitor → improve.
- These are high-level diagrams inferred from public sources, not official internal workflows.



# 9. Comparison Table

- Instagram: social sharing/discovery; examples emphasize recommendation models and safe launches.
- Spotify: audio streaming/discovery; examples emphasize cross-functional teams and developer platforms.
- Both require specification, implementation, validation, maintenance, and feedback.
- Both appear iterative in the examples; public sources do not establish a single model for all teams.

# 10. Waterfall vs Agile & Conclusion

- Waterfall suits stable requirements and formal documentation.
- Agile/incremental suits changing needs and frequent feedback.
- Large projects can combine approaches.
- I learned there is no single best process; it depends on the product, risk, users, and release needs.

# 11. References


- Meta Engineering (2025): Instagram recommendation system and model launches.
- Meta AI (2019): Instagram Explore recommender system.
- Spotify Engineering (2021/2026): Backstage and developer effectiveness.
- Spotify Technology (2026): cross-functional teams and development tools.